Newton-Cotes Formulas of Closed Type

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Simpson’s 3/8 Rule

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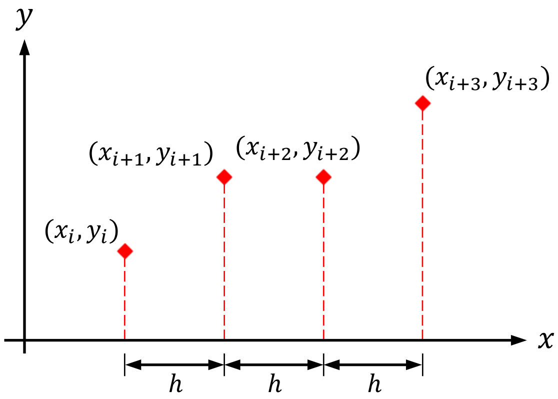
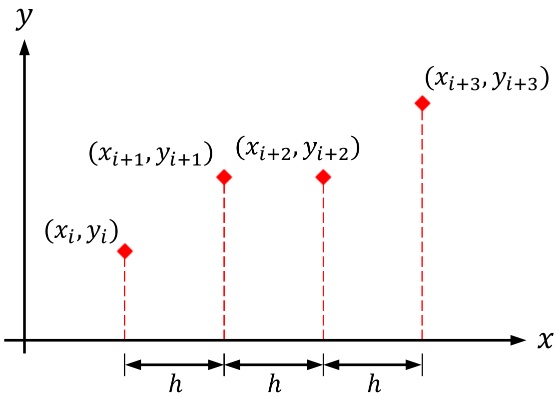


Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule

$n = 3$ $f(x) \approx p_3(x)$

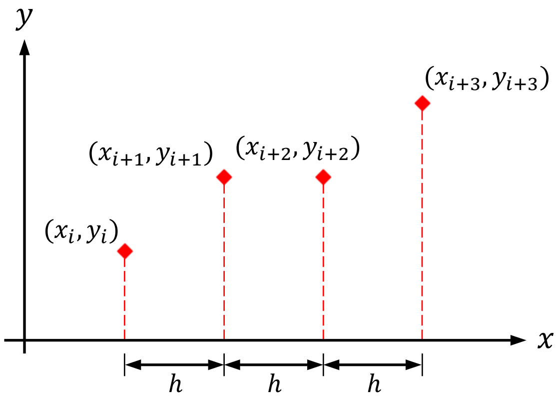


Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule

$n = 3$

$$f(x) \approx p_3(x)$$
$$= L_0^{(3)}(x) \cdot f(x_i) + L_1^{(3)}(x) \cdot f(x_{i+1}) + L_2^{(3)}(x) \cdot f(x_{i+2}) + L_3^{(3)}(x) \cdot f(x_{i+3})$$



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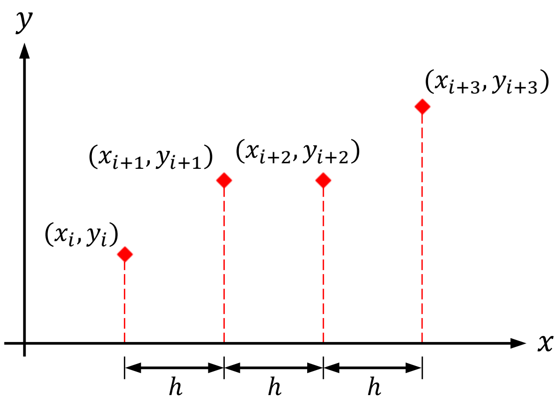
$$= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_i-x_{i+3}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_{i+1}-x_{i+3}} \cdot f(x_{i+1})$$

$$+ \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot \frac{x-x_{i+3}}{x_{i+2}-x_{i+3}} \cdot f(x_{i+2}) + \frac{x-x_i}{x_{i+3}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+3}-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_{i+3}-x_{i+2}} \cdot f(x_{i+3})$$



Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule

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$$
\begin{aligned}
f(x) &\approx p_3(x) \\
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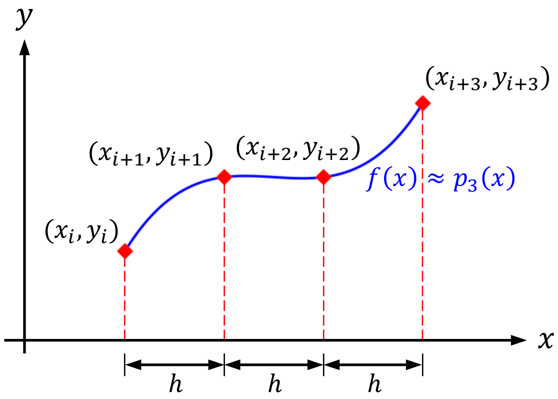
&= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_i-x_{i+3}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_{i+1}-x_{i+3}} \cdot f(x_{i+1}) \\
&\quad + \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot \frac{x-x_{i+3}}{x_{i+2}-x_{i+3}} \cdot f(x_{i+2}) + \frac{x-x_i}{x_{i+3}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+3}-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_{i+3}-x_{i+2}} \cdot f(x_{i+3})
\end{aligned}
$$



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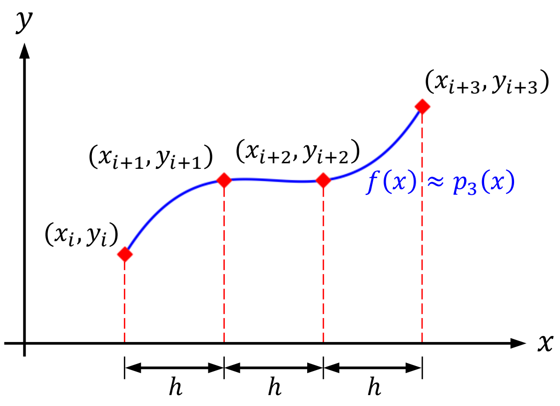
$$+ \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot \frac{x-x_{i+3}}{x_{i+2}-x_{i+3}} \cdot f(x_{i+2}) + \frac{x-x_i}{x_{i+3}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+3}-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_{i+3}-x_{i+2}} \cdot f(x_{i+3})$$

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx$$



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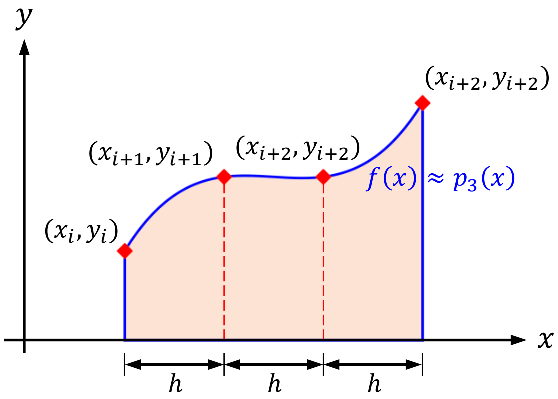

Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule

$\boxed{n = 3}$

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\begin{aligned}
f(x) &\approx p_3(x) \\
&= L_0^{(3)}(x) \cdot f(x_i) + L_1^{(3)}(x) \cdot f(x_{i+1}) + L_2^{(3)}(x) \cdot f(x_{i+2}) + L_3^{(3)}(x) \cdot f(x_{i+3}) \\
&= \tfrac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \tfrac{x-x_{i+2}}{x_i-x_{i+2}} \cdot \tfrac{x-x_{i+3}}{x_i-x_{i+3}} \cdot f(x_i) + \tfrac{x-x_i}{x_{i+1}-x_i} \cdot \tfrac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot \tfrac{x-x_{i+3}}{x_{i+1}-x_{i+3}} \cdot f(x_{i+1}) \\
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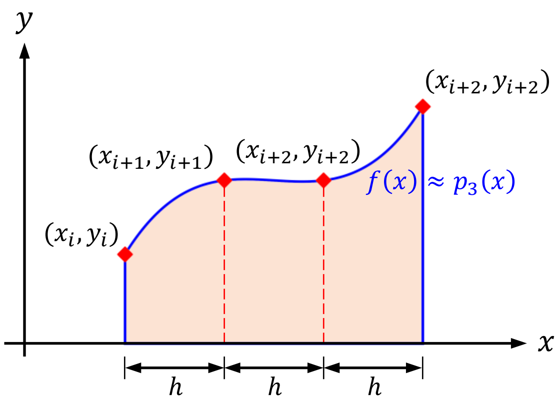
$$



$$
\int_{x_i}^{x_{i+3}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx
$$

$$
\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h}
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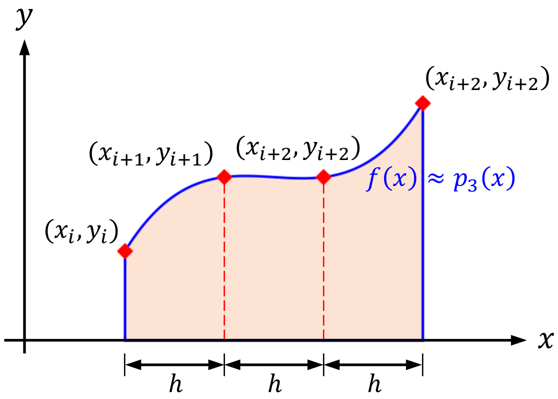

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx$$

$$\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h} \quad \rightarrow \quad d\xi = \frac{1}{h} \cdot dx$$

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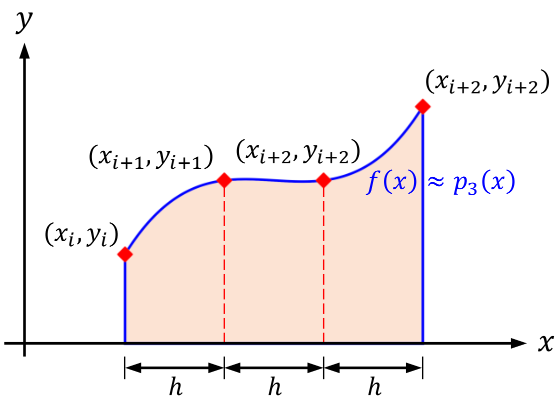

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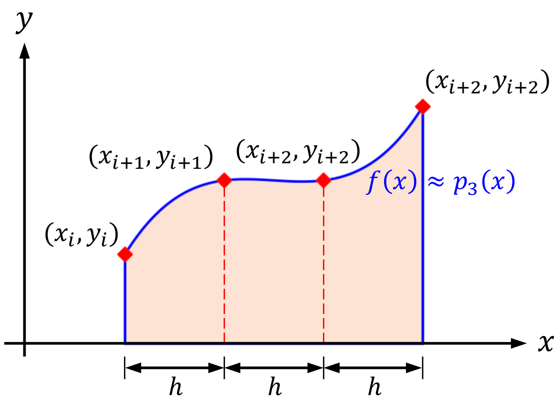
&= \frac{x-x_{i+1}}{x_i-x_{i+1}}\cdot\frac{x-x_{i+2}}{x_i-x_{i+2}}\cdot\frac{x-x_{i+3}}{x_i-x_{i+3}}\cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i}\cdot\frac{x-x_{i+2}}{x_{i+1}-x_{i+2}}\cdot\frac{x-x_{i+3}}{x_{i+1}-x_{i+3}}\cdot f(x_{i+1}) \\
&\quad + \frac{x-x_i}{x_{i+2}-x_i}\cdot\frac{x-x_{i+1}}{x_{i+2}-x_{i+1}}\cdot\frac{x-x_{i+3}}{x_{i+2}-x_{i+3}}\cdot f(x_{i+2}) + \frac{x-x_i}{x_{i+3}-x_i}\cdot\frac{x-x_{i+1}}{x_{i+3}-x_{i+1}}\cdot\frac{x-x_{i+2}}{x_{i+3}-x_{i+2}}\cdot f(x_{i+3})
\end{aligned}
$$



$$\int_{x_i}^{x_{i+3}} f(x)\cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x)\cdot dx$$

$$\xi = \frac{x-x_i}{x_{i+1}-x_i} = \frac{x-x_i}{h} \quad\rightarrow\quad d\xi = \frac{1}{h}\cdot dx$$

$$
\begin{aligned}
x = x_i &\rightarrow \xi = 0 \\
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x = x_{i+3} &\rightarrow \xi = 3
\end{aligned}
$$

$$\int_{x_i}^{x_{i+3}} \frac{x-x_{i+1}}{x_i-x_{i+1}}\cdot\frac{x-x_{i+2}}{x_i-x_{i+2}}\cdot\frac{x-x_{i+3}}{x_i-x_{i+3}}\cdot dx = \int_0^3 \frac{h\cdot(\xi-1)}{-h}\cdot\frac{h\cdot(\xi-2)}{-2h}\cdot\frac{h\cdot(\xi-3)}{-3h}\cdot(h\cdot d\xi) = \frac{3h}{8}$$

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Simpson's 3/8 Rule

$\boxed{n = 3}$

$$f(x) \approx p_3(x)$$
$$= L_0^{(3)}(x) \cdot f(x_i) + L_1^{(3)}(x) \cdot f(x_{i+1}) + L_2^{(3)}(x) \cdot f(x_{i+2}) + L_3^{(3)}(x) \cdot f(x_{i+3})$$
$$= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_i-x_{i+3}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_{i+1}-x_{i+3}} \cdot f(x_{i+1})$$
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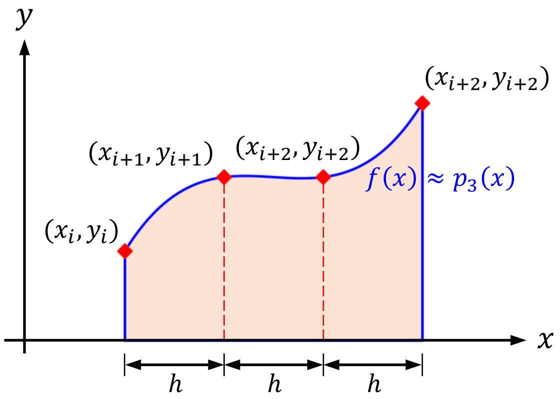

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx$$

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$x = x_{i+3} \quad \rightarrow \quad \xi = 3$

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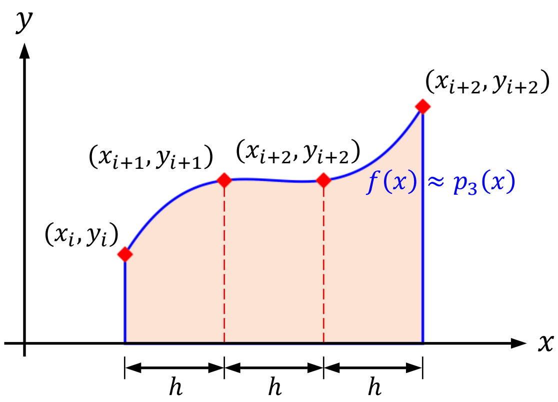

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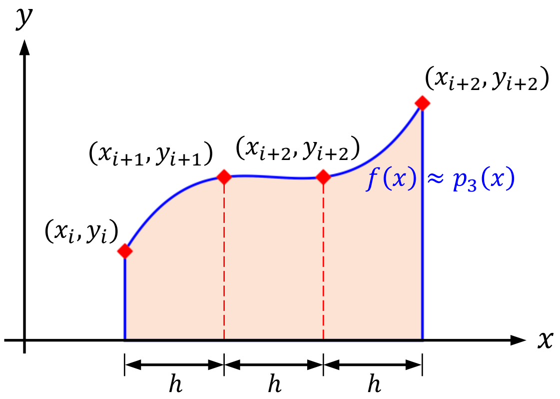


$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx$$

$$\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h} \rightarrow d\xi = \frac{1}{h} \cdot dx$$

$x = x_i \rightarrow \xi = 0$
$x = x_{i+1} \rightarrow \xi = 1$
$x = x_{i+2} \rightarrow \xi = 2$
$x = x_{i+3} \rightarrow \xi = 3$

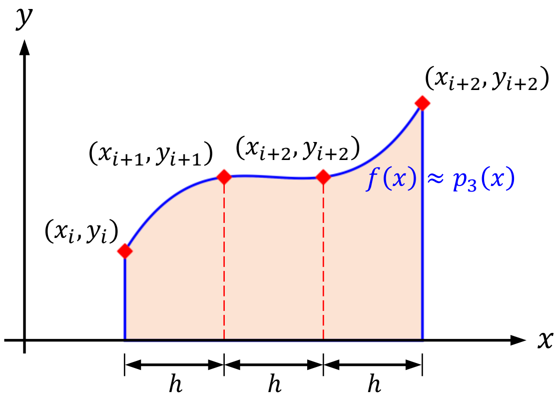
$$\int_{x_i}^{x_{i+3}} \frac{x - x_i}{x_{i+2} - x_i} \cdot \frac{x - x_{i+1}}{x_{i+2} - x_{i+1}} \cdot \frac{x - x_{i+3}}{x_{i+2} - x_{i+3}} \cdot dx = \int_0^3 \frac{h \cdot (\xi)}{2h} \cdot \frac{h \cdot (\xi - 1)}{h} \cdot \frac{h \cdot (\xi - 3)}{-h} \cdot (h \cdot d\xi) = \frac{9h}{8}$$

Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule

$\boxed{n = 3}$

$$\begin{aligned} f(x) &\approx p_3(x) \\ &= L_0^{(3)}(x) \cdot f(x_i) + L_1^{(3)}(x) \cdot f(x_{i+1}) + L_2^{(3)}(x) \cdot f(x_{i+2}) + L_3^{(3)}(x) \cdot f(x_{i+3}) \\ &= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_i-x_{i+3}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_{i+1}-x_{i+3}} \cdot f(x_{i+1}) \\ &\quad + \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot \frac{x-x_{i+3}}{x_{i+2}-x_{i+3}} \cdot f(x_{i+2}) + \frac{x-x_i}{x_{i+3}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+3}-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_{i+3}-x_{i+2}} \cdot f(x_{i+3}) \end{aligned}$$



$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx$$

$$\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h} \quad \rightarrow \quad d\xi = \frac{1}{h} \cdot dx$$

$$\begin{aligned} x = x_i \quad &\rightarrow \quad \xi = 0 \\ x = x_{i+1} \quad &\rightarrow \quad \xi = 1 \\ x = x_{i+2} \quad &\rightarrow \quad \xi = 2 \\ x = x_{i+3} \quad &\rightarrow \quad \xi = 3 \end{aligned}$$

$$\int_{x_i}^{x_{i+3}} \frac{x - x_i}{x_{i+2} - x_i} \cdot \frac{x - x_{i+1}}{x_{i+2} - x_{i+1}} \cdot \frac{x - x_{i+3}}{x_{i+2} - x_{i+3}} \cdot dx = \int_0^3 \frac{h \cdot (\xi)}{2h} \cdot \frac{h \cdot (\xi - 1)}{h} \cdot \frac{h \cdot (\xi - 3)}{-h} \cdot (h \cdot d\xi) = \frac{9h}{8}$$

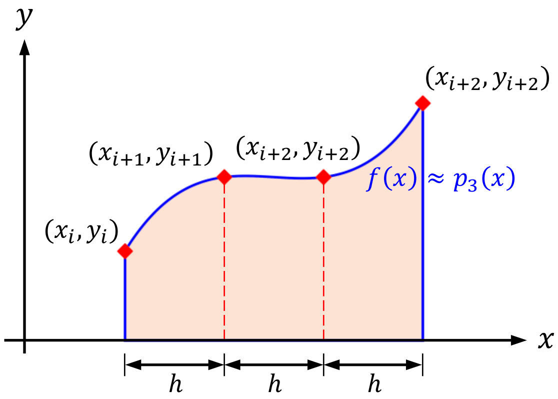
$$\int_{x_i}^{x_{i+3}} \frac{x - x_i}{x_{i+3} - x_i} \cdot \frac{x - x_{i+1}}{x_{i+3} - x_{i+1}} \cdot \frac{x - x_{i+2}}{x_{i+3} - x_{i+2}} \cdot dx = \int_0^3 \frac{h \cdot (\xi)}{3h} \cdot \frac{h \cdot (\xi - 1)}{2h} \cdot \frac{h \cdot (\xi - 2)}{h} \cdot (h \cdot d\xi) = \frac{3h}{8}$$

Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule

$n = 3$

$$f(x) \approx p_3(x)$$
$$= L_0^{(3)}(x) \cdot f(x_i) + L_1^{(3)}(x) \cdot f(x_{i+1}) + L_2^{(3)}(x) \cdot f(x_{i+2}) + L_3^{(3)}(x) \cdot f(x_{i+3})$$
$$= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_i-x_{i+3}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot \frac{x-x_{i+3}}{x_{i+1}-x_{i+3}} \cdot f(x_{i+1})$$
$$+ \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot \frac{x-x_{i+3}}{x_{i+2}-x_{i+3}} \cdot f(x_{i+2}) + \frac{x-x_i}{x_{i+3}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+3}-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_{i+3}-x_{i+2}} \cdot f(x_{i+3})$$



$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx = \frac{3h}{8} \cdot [f(x_i) + 3 \cdot f(x_{i+1}) + 3 \cdot f(x_{i+2}) + f(x_{i+3})]$$

$$\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h} \quad \rightarrow \quad d\xi = \frac{1}{h} \cdot dx$$

$x = x_i \quad \rightarrow \quad \xi = 0$
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$x = x_{i+3} \quad \rightarrow \quad \xi = 3$

$$\int_{x_i}^{x_{i+3}} \frac{x - x_i}{x_{i+2} - x_i} \cdot \frac{x - x_{i+1}}{x_{i+2} - x_{i+1}} \cdot \frac{x - x_{i+3}}{x_{i+2} - x_{i+3}} \cdot dx = \int_0^3 \frac{h \cdot (\xi)}{2h} \cdot \frac{h \cdot (\xi - 1)}{h} \cdot \frac{h \cdot (\xi - 3)}{-h} \cdot (h \cdot d\xi) = \frac{9h}{8}$$

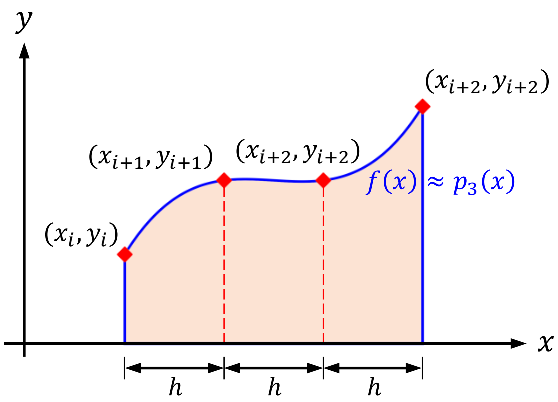
$$\int_{x_i}^{x_{i+3}} \frac{x - x_i}{x_{i+3} - x_i} \cdot \frac{x - x_{i+1}}{x_{i+3} - x_{i+1}} \cdot \frac{x - x_{i+2}}{x_{i+3} - x_{i+2}} \cdot dx = \int_0^3 \frac{h \cdot (\xi)}{3h} \cdot \frac{h \cdot (\xi - 1)}{2h} \cdot \frac{h \cdot (\xi - 2)}{h} \cdot (h \cdot d\xi) = \frac{3h}{8}$$

Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule

$\boxed{n = 3}$

$$f(x) \approx p_3(x)$$
$$= L_0^{(3)}(x)\cdot f(x_i) + L_1^{(3)}(x)\cdot f(x_{i+1}) + L_2^{(3)}(x)\cdot f(x_{i+2}) + L_3^{(3)}(x)\cdot f(x_{i+3})$$
$$= \frac{x-x_{i+1}}{x_i-x_{i+1}}\cdot\frac{x-x_{i+2}}{x_i-x_{i+2}}\cdot\frac{x-x_{i+3}}{x_i-x_{i+3}}\cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i}\cdot\frac{x-x_{i+2}}{x_{i+1}-x_{i+2}}\cdot\frac{x-x_{i+3}}{x_{i+1}-x_{i+3}}\cdot f(x_{i+1})$$
$$+ \frac{x-x_i}{x_{i+2}-x_i}\cdot\frac{x-x_{i+1}}{x_{i+2}-x_{i+1}}\cdot\frac{x-x_{i+3}}{x_{i+2}-x_{i+3}}\cdot f(x_{i+2}) + \frac{x-x_i}{x_{i+3}-x_i}\cdot\frac{x-x_{i+1}}{x_{i+3}-x_{i+1}}\cdot\frac{x-x_{i+2}}{x_{i+3}-x_{i+2}}\cdot f(x_{i+3})$$



$$\boxed{\int_{x_i}^{x_{i+3}} f(x)\cdot dx \approx \int_{x_i}^{x_{i+3}} p_3(x)\cdot dx = \frac{3h}{8}\cdot\left[f(x_i) + 3\cdot f(x_{i+1}) + 3\cdot f(x_{i+2}) + f(x_{i+3})\right]}$$

$$\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h} \quad\rightarrow\quad d\xi = \frac{1}{h}\cdot dx$$

$$x = x_i \quad\rightarrow\quad \xi = 0$$
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$$x = x_{i+3} \quad\rightarrow\quad \xi = 3$$

$$\int_{x_i}^{x_{i+3}} \frac{x - x_i}{x_{i+2} - x_i}\cdot\frac{x - x_{i+1}}{x_{i+2} - x_{i+1}}\cdot\frac{x - x_{i+3}}{x_{i+2} - x_{i+3}}\cdot dx = \int_0^3 \frac{h\cdot(\xi)}{2h}\cdot\frac{h\cdot(\xi-1)}{h}\cdot\frac{h\cdot(\xi-3)}{-h}\cdot(h\cdot d\xi) = \frac{9h}{8}$$

$$\int_{x_i}^{x_{i+3}} \frac{x - x_i}{x_{i+3} - x_i}\cdot\frac{x - x_{i+1}}{x_{i+3} - x_{i+1}}\cdot\frac{x - x_{i+2}}{x_{i+3} - x_{i+2}}\cdot dx = \int_0^3 \frac{h\cdot(\xi)}{3h}\cdot\frac{h\cdot(\xi-1)}{2h}\cdot\frac{h\cdot(\xi-2)}{h}\cdot(h\cdot d\xi) = \frac{3h}{8}$$

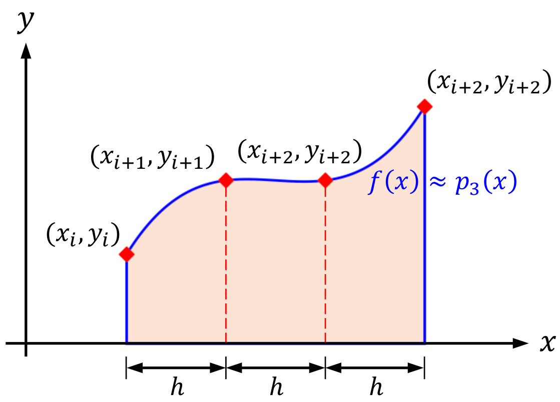
Newton-Cotes Formulas of Closed Type

Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

Newton-Cotes Formulas of Closed Type

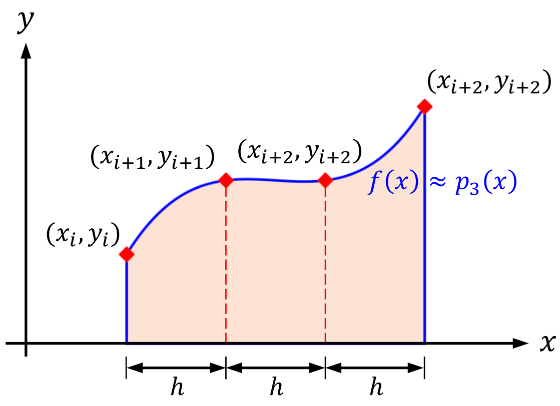
Simpson's 3/8 Rule: The Local Truncation Error



Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \cdots$$



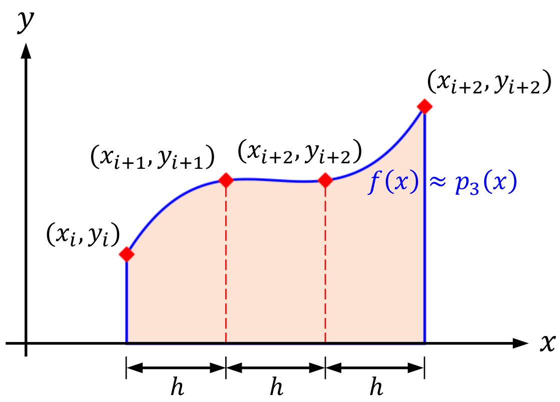
Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \cdots$$

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx = (x_{i+3} - x_i) \cdot f(x_i) + \frac{(x_{i+3} - x_i)^2}{2!} \cdot f'(x_i)$$

$$+ \frac{(x_{i+3} - x_i)^3}{3!} \cdot f''(x_i) + \frac{(x_{i+3} - x_i)^4}{4!} \cdot f^{(3)}(x_i) + \cdots$$



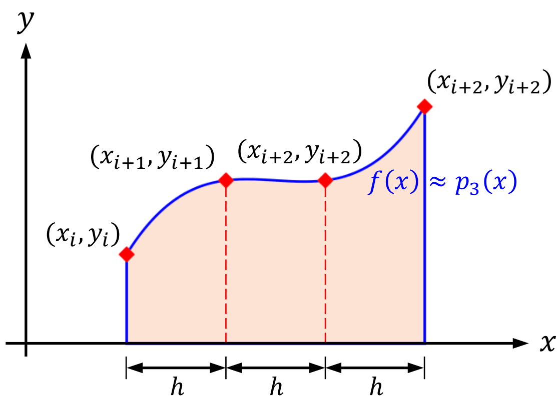
Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \cdots$$

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx = (3h) \cdot f(x_i) + \frac{(3h)^2}{2} \cdot f'(x_i)$$

$$+ \frac{(3h)^3}{6} \cdot f''(x_i) + \frac{(3h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(3h)^5}{120} \cdot f^{(4)}(x_i) + \cdots$$



Newton-Cotes Formulas of Closed Type

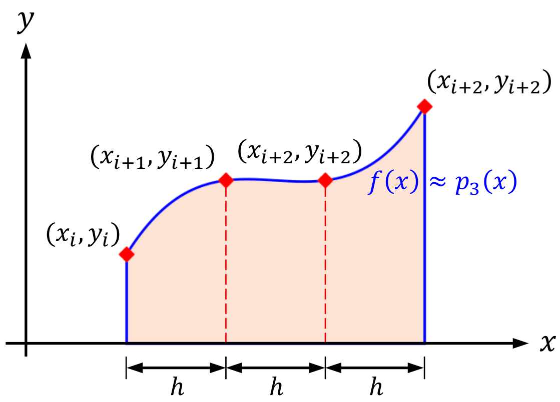
Simpson's 3/8 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \cdots$$

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx = (3h) \cdot f(x_i) + \frac{(3h)^2}{2} \cdot f'(x_i)$$

$$+ \frac{(3h)^3}{6} \cdot f''(x_i) + \frac{(3h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(3h)^5}{120} \cdot f^{(4)}(x_i) + \cdots$$

$$R^{(3)}_{[x_i, x_{i+3}]} = \int_{x_i}^{x_{i+3}} f(x) \cdot dx - \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx$$

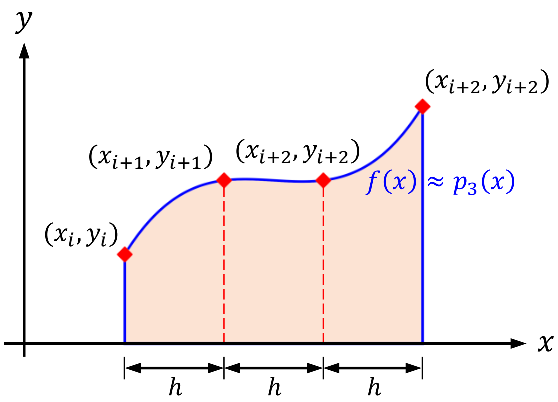


Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i)\cdot f'(x_i) + \frac{(x-x_i)^2}{2!}\cdot f''(x_i) + \frac{(x-x_i)^3}{3!}\cdot f^{(3)}(x_i) + \cdots$$

$$\int_{x_i}^{x_{i+3}} f(x)\cdot dx = (3h)\cdot f(x_i) + \frac{(3h)^2}{2}\cdot f'(x_i) + \frac{(3h)^3}{6}\cdot f''(x_i) + \frac{(3h)^4}{24}\cdot f^{(3)}(x_i) + \frac{(3h)^5}{120}\cdot f^{(4)}(x_i) + \cdots$$



$$R^{(3)}_{[x_i,x_{i+3}]} = \int_{x_i}^{x_{i+3}} f(x)\cdot dx - \int_{x_i}^{x_{i+3}} p_3(x)\cdot dx = 3h\cdot f(x_i) + \frac{9h^2}{2}\cdot f'(x_i) + \frac{9h^3}{2}\cdot f''(x_i) + \frac{27h^4}{8}\cdot f^{(3)}(x_i) + \frac{81h^5}{40}\cdot f^{(4)}(x_i) + \cdots$$

$$- \frac{3h}{8}\cdot f(x_i) - \frac{9h}{8}\cdot f(x_{i+1}) - \frac{9h}{8}\cdot f(x_{i+2}) - \frac{3h}{8}\cdot f(x_{i+3})$$

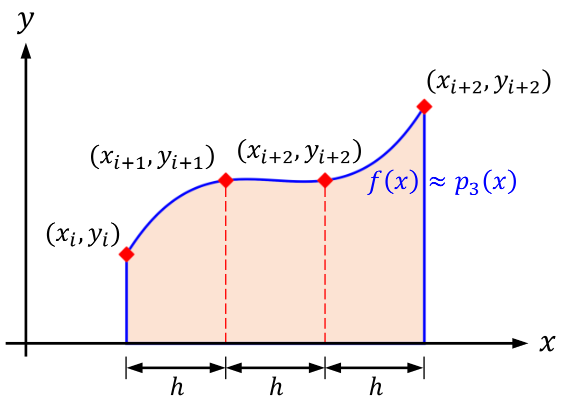
Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \cdots$$

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx = (3h) \cdot f(x_i) + \frac{(3h)^2}{2} \cdot f'(x_i)$$

$$+ \frac{(3h)^3}{6} \cdot f''(x_i) + \frac{(3h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(3h)^5}{120} \cdot f^{(4)}(x_i) + \cdots$$



$$R^{(3)}_{[x_i, x_{i+3}]} = \int_{x_i}^{x_{i+3}} f(x) \cdot dx - \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx = 3h \cdot f(x_i) + \frac{9h^2}{2} \cdot f'(x_i) + \frac{9h^3}{2} \cdot f''(x_i) + \frac{27h^4}{8} \cdot f^{(3)}(x_i) + \frac{81h^5}{40} \cdot f^{(4)}(x_i) + \cdots$$

$$- \frac{3h}{8} \cdot f(x_i) - \frac{9h}{8} \cdot \left[f(x_i) + h \cdot f'(x_i) + \frac{h^2}{2} \cdot f''(x_i) + \frac{h^3}{6} \cdot f^{(3)}(x_i) + \frac{h^4}{24} \cdot f^{(4)}(x_i) + \cdots \right]$$

$$- \frac{9h}{8} \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2} \cdot f''(x_i) + \frac{(2h)^3}{6} \cdot f^{(3)}(x_i) + \frac{(2h)^4}{24} \cdot f^{(4)}(x_i) + \cdots \right]$$

$$- \frac{3h}{8} \cdot \left[f(x_i) + (3h) \cdot f'(x_i) + \frac{(3h)^2}{2} \cdot f''(x_i) + \frac{(3h)^3}{6} \cdot f^{(3)}(x_i) + \frac{(3h)^4}{24} \cdot f^{(4)}(x_i) + \cdots \right]$$

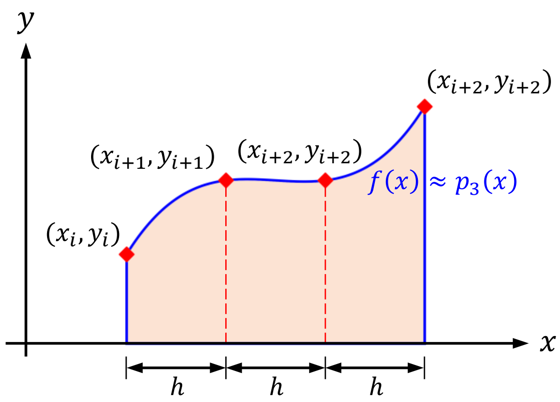
Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \cdots$$

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx = (3h) \cdot f(x_i) + \frac{(3h)^2}{2} \cdot f'(x_i)$$

$$+ \frac{(3h)^3}{6} \cdot f''(x_i) + \frac{(3h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(3h)^5}{120} \cdot f^{(4)}(x_i) + \cdots$$



$$R^{(3)}_{[x_i, x_{i+3}]} = \int_{x_i}^{x_{i+3}} f(x) \cdot dx - \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx = 3h \cdot f(x_i) + \frac{9h^2}{2} \cdot f'(x_i) + \frac{9h^3}{2} \cdot f''(x_i) + \frac{27h^4}{8} \cdot f^{(3)}(x_i) + \frac{81h^5}{40} \cdot f^{(4)}(x_i) + \cdots$$

$$- 3h \cdot f(x_i) - \frac{9h^2}{2} \cdot f'(x_i) - \frac{9h^3}{2} \cdot f''(x_i) - \frac{27h^4}{8} \cdot f^{(3)}(x_i) - \frac{33h^5}{16} \cdot f^{(4)}(x_i) - \cdots$$

Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \cdots$$

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx = (3h) \cdot f(x_i) + \frac{(3h)^2}{2} \cdot f'(x_i)$$

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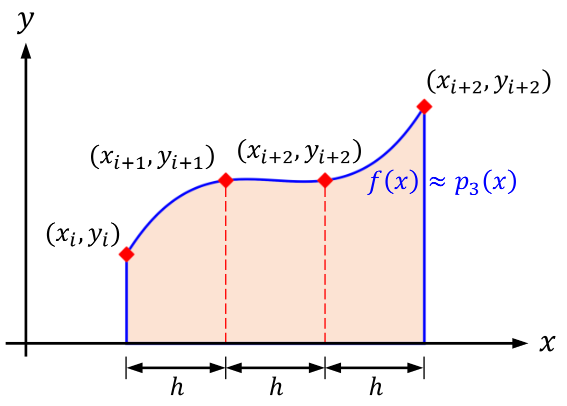
$$R^{(3)}_{[x_i, x_{i+3}]} = \int_{x_i}^{x_{i+3}} f(x) \cdot dx - \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx = -\frac{3h^5}{80} \cdot f^{(4)}(x_i) + \cdots$$



Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule: The Local Truncation Error

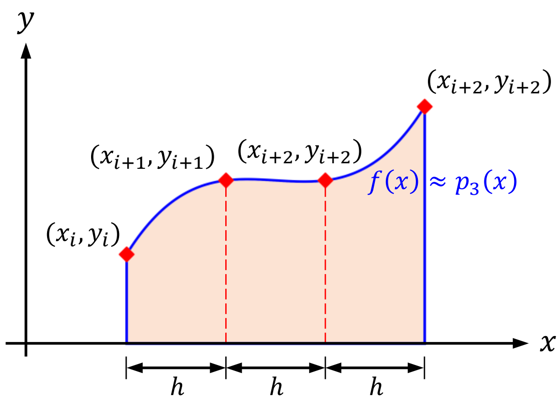
$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \cdots$$

$$\int_{x_i}^{x_{i+3}} f(x) \cdot dx = (3h) \cdot f(x_i) + \frac{(3h)^2}{2} \cdot f'(x_i)$$

$$+ \frac{(3h)^3}{6} \cdot f''(x_i) + \frac{(3h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(3h)^5}{120} \cdot f^{(4)}(x_i) + \cdots$$

$$R^{(3)}_{[x_i, x_{i+3}]} = \int_{x_i}^{x_{i+3}} f(x) \cdot dx - \int_{x_i}^{x_{i+3}} p_3(x) \cdot dx = -\frac{3h^5}{80} \cdot f^{(4)}(x_i) + \cdots$$

$$= -\frac{3h^5}{80} \cdot f^{(4)}(\xi_{i+1}); \quad x_i \le \xi_{i+1} \le x_{i+3}$$



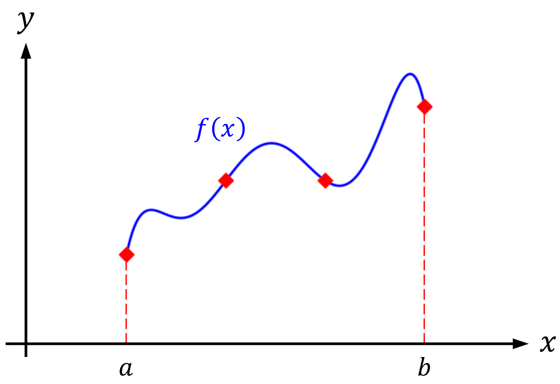
Newton-Cotes Formulas of Closed Type

Newton-Cotes Formulas of Closed Type

Simpson’s 3/8 Rule : The Global Truncation Error

Newton-Cotes Formulas of Closed Type

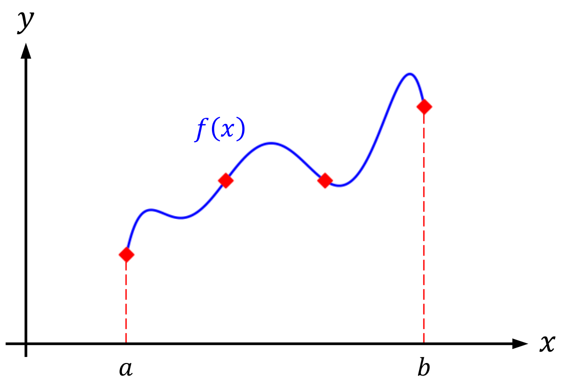
Simpson's 3/8 Rule : The Global Truncation Error



Newton-Cotes Formulas of Closed Type

Simpson's 3/8 Rule : The Global Truncation Error

$$\int_a^b f(x) \cdot dx$$



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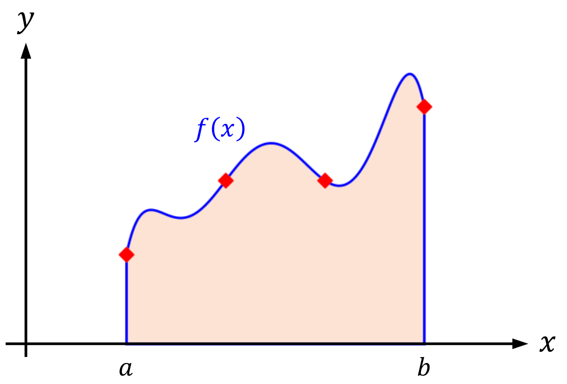

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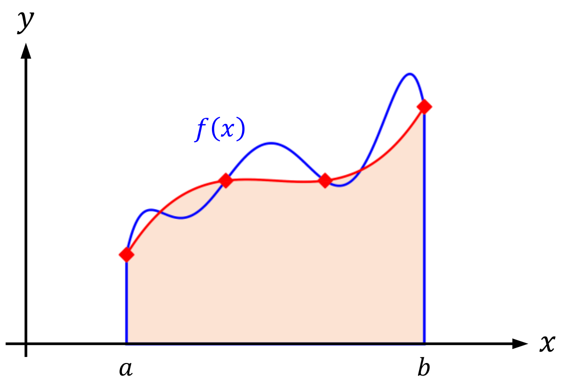

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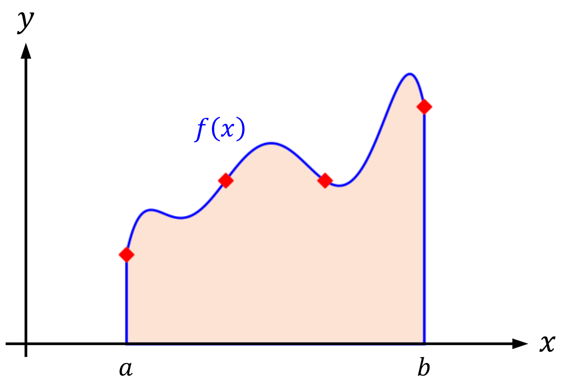

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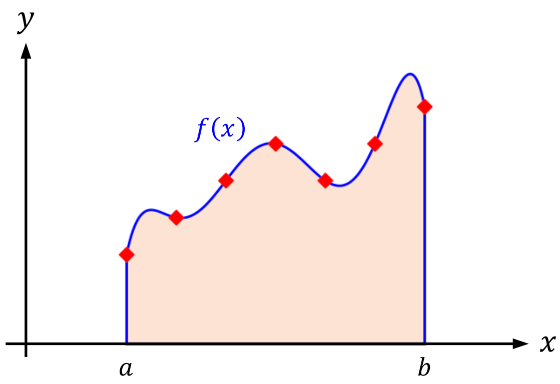

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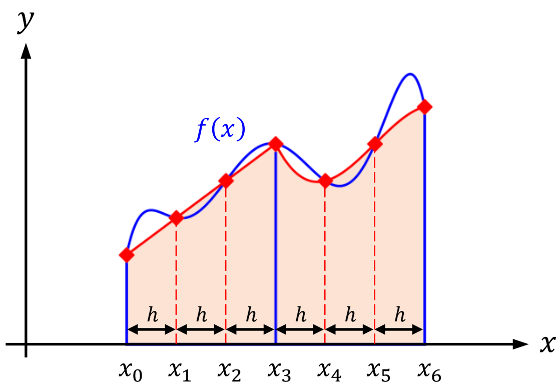

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$$\int_a^b f(x)\cdot dx = \int_a^{a+3h} f(x)\cdot dx + \int_{a+3h}^{a+6h} f(x)\cdot dx + \cdots + \int_{b-3h}^{b-3h} f(x)\cdot dx + \int_{b-3h}^{b} f(x)\cdot dx$$

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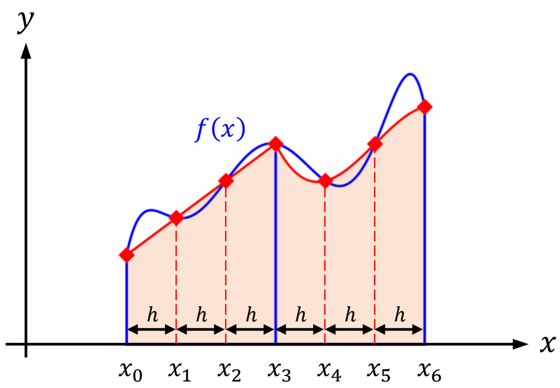

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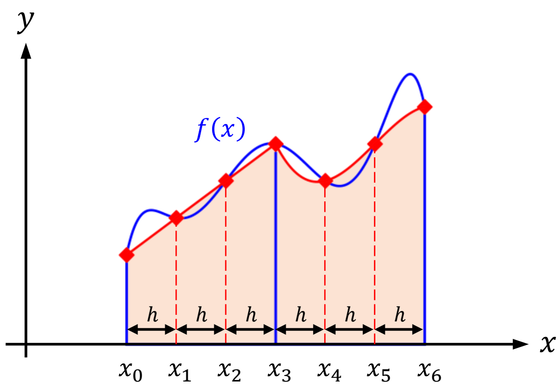

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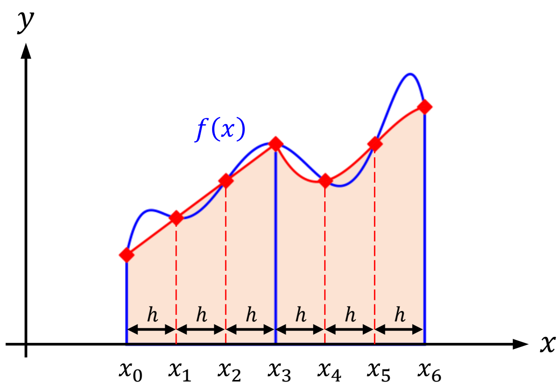

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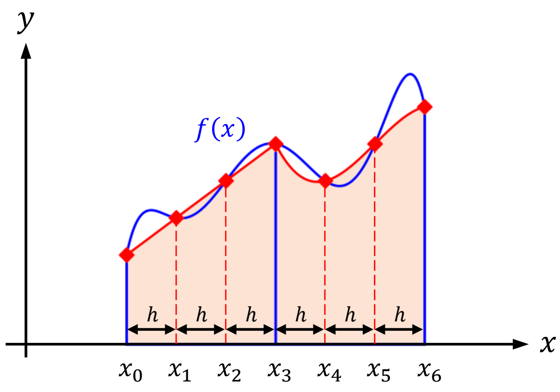
$$R^{(3)}_{[a,b]} = R^{(3)}_{[x_0,x_3]} + R^{(3)}_{[x_3,x_6]} + \cdots + R^{(3)}_{[x_{N-3},x_N]}$$



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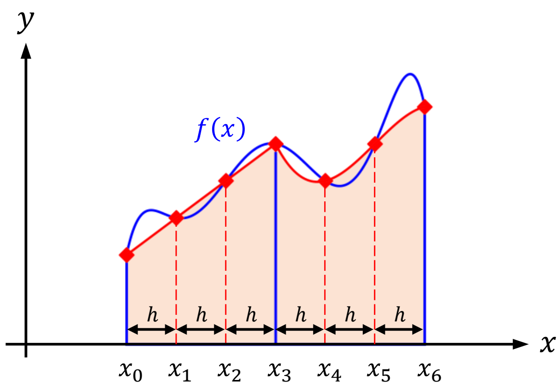

$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b-a}{N} = x_1 - x_0 = x_2 - x_1 = \cdots = x_N - x_{N-1}$$

$$R_{[a,b]}^{(3)} = R_{[x_0,x_3]}^{(3)} + R_{[x_3,x_6]}^{(3)} + \cdots + R_{[x_{N-3},x_N]}^{(3)} = -\frac{3h^5}{80}\cdot f^{(4)}(\xi_1) - \frac{3h^5}{80}\cdot f^{(4)}(\xi_2) - \cdots - \frac{3h^5}{80}\cdot f^{(4)}(\xi_{N/3});$$

$$\begin{gathered} x_0 \le \xi_1 \le x_3 \\ x_3 \le \xi_2 \le x_6 \\ \vdots \\ x_{N-3} \le \xi_{N/3} \le x_N \end{gathered}$$

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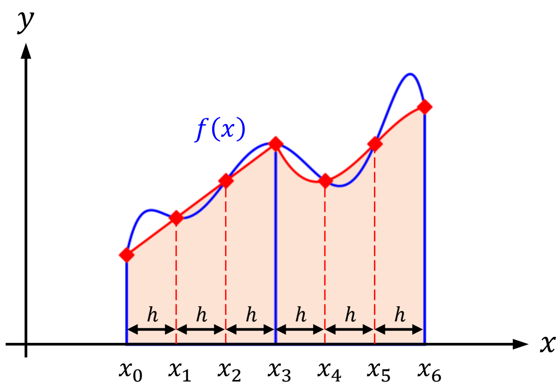

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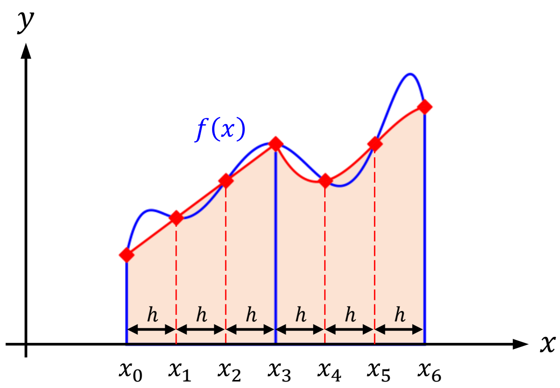

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$$x_3 \le \xi_2 \le x_6$$
$$\vdots$$
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$$= -\frac{\frac{b-a}{h}\cdot 3h^5}{3\cdot 80}\cdot f^{(4)}(\xi); \quad x_0 \le \xi \le x_N$$

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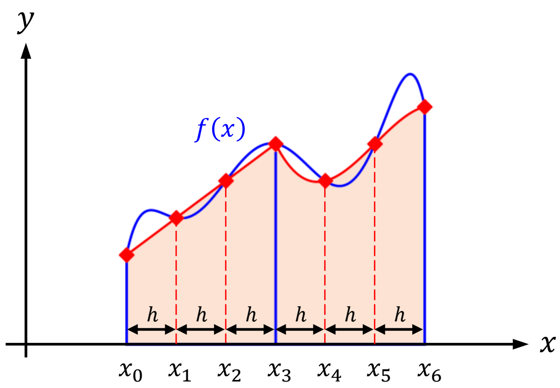

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